

Contact

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Top Skills

SQL

Smoothed particle hydrodynamics
(SPH)

Data Analysis

Languages

English (Professional Working)

Italian (Native or Bilingual)

German (Elementary)

Spanish (Professional Working)

Certifications

MATLAB & Simulink – An
Introduction

Introduction to FEA with MSC Apex
(Academic Version)

Time Management Fundamentals

Certificate of attendance: Fatigue and
Fracture in FEA

Building Focus in a Distracting World

Honors-Awards

Mobilità Internazionale

Erasmus+ Scholarship

Erasmus+ Scholarship

Challenge ASI02-Actinspace 2020

PhD Scholarship

Publications

The role of intra-laminar
hybridization on the thermo-
mechanical behaviour of fibre-hybrid
composites using 3D RVEs

Elastic Properties Prediction in Yarn-
level Hybrid Composite Lamina

Giuseppe Romano, PhD, MIMMM, CEng MIMechE, FHEA

CAE & Materials Engineer | STEM Mentor & Educator | PhD,
MIMMM, CEng MIMechE, FHEA
Cambridge, England, United Kingdom

Summary

I'm a PhD-qualified Mechanical Engineer working at the intersection of materials science, simulation, and digital engineering. Currently, I'm an R&D Engineer II at Ansys, where I contribute to the development of material intelligence tools and data-driven workflows across the Ansys simulation platform. My work focuses on enhancing how engineers and designers interact with materials data to solve real-world challenges across industries like aerospace, automotive, and energy.

My academic background includes a PhD from the University of Manchester, where I developed micromechanical models for fibre-hybrid composites—integrating computational analysis, laminate manufacturing, and experimental validation. I also hold an MSc in Mechanical Engineering from the University of Calabria, with a focus on residual stresses in thermoset composites. My international experience spans Erasmus+ research exchanges in Spain and Germany, enriching my global perspective on engineering and collaboration.

As the Italian Regional Representative for NAFEMS, I helped promoting simulation best practices through events, webinars, and community building. I'm also the founder of Make It STEM, a personal initiative to support STEM students and professionals through career-focused content and online courses.

My technical toolkit includes:

Finite Element Analysis (FEA)

Materials Informatics

Python & Data Processing

Composite Materials & Manufacturing

MICROMECHANICS OF INTRA-LAMINAR HYBRID LAMINA WITH HOLLOW FIBRES: A RVE MODEL

Development of a Novel Spinneret Design for Improved Melt Extrusion Performance: A Computational and Empirical Study

On the Micro-Stress Fields in Fibre-Hybrid Polymer Composite Laminae with Periodic Microstructure Under Transverse Tension and Transverse Shear Loading

CAE Software Integration

I'm passionate about bridging the gap between research and application, mentoring the next generation of engineers, and using simulation to accelerate innovation. Let's connect if you're working in CAE, materials R&D, or STEM education.

Experience

Synopsys Inc

1 year

R&D Engineering, Senior Engineer

January 2026 - Present (6 months)

Cambridge, England, United Kingdom

R&D Engineer II | Ansys-Materials (Data Products)

July 2025 - January 2026 (7 months)

Cambridge, England, United Kingdom

The University of Manchester

6 years 9 months

Visiting Researcher

August 2025 - Present (11 months)

Visiting researcher at the Department of Mechanical and Aerospace Engineering, School of Engineering

Teaching Assistant

October 2019 - July 2025 (5 years 10 months)

Manchester, England, United Kingdom

Teaching Assistant for the following courses:

-AERO21111 Space Systems, 2023-24 1st Semester

-AERO10301 Aerospace Design, 2022-23 1st Semester, 2023-24 1st Semester

-AERO20062&20092 MACE20062-20092 Modelling & Simulation 2, 2022-23 2nd Semester

-MACE10302 Introduction to Aircraft Design, 2021-22 2nd Semester

-MACE12101&12301 Tools for Engineers (Aerospace and Mechanical) 2020-21 1st Semester, 2021-22 1st Semester, 2022-23 2nd Semester

-MACE31302 Structures 3 (Aerospace) 2020-21 2nd Semester, 2021-22 2nd Semester, 2022-23 2nd Semester

I have been involved in marking, lab tutorials and computer sessions.

Visiting Research Associate

December 2023 - December 2024 (1 year 1 month)

Manchester Area, United Kingdom

The project title is 'Development of simplified approaches for weld residual stress modelling in critical high-temperature components'. The research project is funded by EDF Energy UK.

Translational projects under Modelling and Simulation Centre (MaSC) <https://www.mace.manchester.ac.uk/msc/about/>

Casual Research Assistant

February 2024 - September 2024 (8 months)

Manchester Area, United Kingdom

Polymer extrusion with FET (<https://www.fetuk.com/>)

Make IT STEM

Founder

April 2025 - Present (1 year 3 months)

Cambridge, England, United Kingdom

As the founder of Make IT STEM, I create tailored educational content, courses, and resources to support students and professionals in STEM fields. My mission is to empower people with the skills, confidence, and strategies they need to secure meaningful roles in industry and academia.

I blend my background in CAE and materials engineering, academic mentorship, and content design to develop engaging, outcome-driven learning experiences. Topics I focus on include:

- Career development in STEM
- Engineering design & simulation tools
- Research communication
- Professional & transferable skills
- Job application and interview strategies

Make IT STEM bridges the gap between technical knowledge and employability, helping the next generation of innovators thrive.

MentorMeet

Coach/Mentor

December 2024 - Present (1 year 7 months)

Providing tailored 1:1 mentoring and coaching sessions for BSc, MSc, and PhD students in STEM disciplines.

Mission Study Abroad

Tutor

March 2025 - Present (1 year 4 months)

Supporting Italian students in successfully applying to PhD programs at top universities in the UK. I provide guidance on academic CV writing, research proposals, and tailored cover letters, as well as advice on selecting suitable programs and supervisors. My role also includes mentoring students on navigating the UK academic system and preparing for interviews, with the aim of enhancing their chances of admission to leading institutions.

Outsmart Insight

Associate

December 2024 - December 2025 (1 year 1 month)

Contributing to in-depth profiling of experts in cutting-edge technology fields. My main responsibilities include conducting structured research on academic and industry professionals, analyzing their areas of expertise, project involvement, and scientific output to support strategic intelligence services for global clients.

NAFEMS

5 years 8 months

Italia Regional Representative

March 2021 - August 2025 (4 years 6 months)

Manchester, England, United Kingdom

Conducting events dedicated to Engineering Simulation in Italy, facilitated by a distinguished committee comprising experts from both academia and industry (https://www.nafems.org/community/regional/italia/steering_committee/); events organised:

- The Italian webinar series <https://www.nafems.org/events/italian-webinar-series/>
- NAFEMS Italia 2024 Regional Conference (<https://www.nafems.org/events/nafems/2024/nafems-italy-conference-2024/>)

Represented NAFEMS at the <https://www.saiebologna.it/it/comunicati/bologna-2022/>

Italian Regional Assistant
January 2020 - February 2021 (1 year 2 months)
Knutsford

Ansys Materials
R&D Engineer II
April 2025 - July 2025 (4 months)
Cambridge, England, United Kingdom

As an R&D Engineer II within the Ansys Granta Materials Data Products Department, I contribute to the development of materials-related capabilities across Ansys' simulation ecosystem. My role bridges materials science and engineering software, supporting the integration of material intelligence into CAE workflows.

Key responsibilities include:

- Contributing to the development and validation of materials data tools and simulation integrations.
- Collaborating cross-functionally with product managers, developers, and application engineers to define and deliver new features.
- Supporting the implementation of materials data strategies across industries such as aerospace, automotive, and energy.

Hexagon Manufacturing Intelligence
Student Ambassador
October 2022 - May 2024 (1 year 8 months)
Manchester Area, United Kingdom

Promoting MSC Apex and Digimat at the University of Manchester

Computer Aided Engineering (CAE) Society - University of Manchester
Co-Founder
February 2023 - March 2024 (1 year 2 months)
Manchester, England, United Kingdom

Actively contributing to the Wing Design Group project by offering guidance and tutorials.

Graphene Innovations Manchester
Research Assistant
April 2022 - September 2023 (1 year 6 months)
Manchester Area, United Kingdom

Developed FEA models, and conducted experiments and tests on graphene-enhanced composites for various applications.

Siemens PLM Software

Master's Thesis Internship

October 2018 - March 2019 (6 months)

Leuven, Flemish Region, Belgium

Title: Numerical assessment of curing residual stresses in thermosetting composite materials in Simcenter 3D

Development of computational micromechanical models (Repeating unit cells and representative volume elements) in Simcenter 3D to predict residual micro-stresses in carbon/epoxy composites resulting from chemical and thermal-induced epoxy resin shrinkage during the curing cycle

Education

The University of Manchester

Doctor of Philosophy - PhD, Mechanical Engineering · (September 2019 - August 2024)

Università degli Studi della Calabria

Master's degree, Mechanical Engineering · (2015 - 2019)

Ostbayerische Technische Hochschule Regensburg

ERASMUS+ Exchange Program, Mechanical Engineering · (March 2017 - July 2017)

Mondragon Unibertsitatea

ERASMUS+ exchange student, Ingeniería industrial · (February 2016 - June 2016)

Università degli Studi della Calabria

Bachelor's degree, Mechanical Engineering · (September 2011 - September 2015)